Crop Diversity as an Insurance Risk Metric

Does Geographic Crop Diversity Predict Agricultural Insurance Loss Ratios?

An Empirical Analysis of USDA Risk Management Agency Data

Research Hypothesis

Counties with higher crop diversity (as measured by Shannon entropy) will exhibit lower insurance loss ratios and lower claim volatility than monoculture-dominant counties, after controlling for exposure size and coverage type.

Data Source	USDA Risk Management Agency (RMA)
Coverage	County-year observations, 1999–2004
Datasets	colsommonth, sobsccc, sobtpu
Pipeline	16-step Python analysis (pandas, statsmodels, scikit-learn)

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Contents

1. Introduction and Motivation

Agricultural insurance is a cornerstone of the US farm safety net, administered through the USDA Risk Management Agency (RMA). Premiums and indemnity payments are determined primarily by historical loss experience and actuarial models that treat each crop independently. A significant open question is whether the *composition* of a county's crop portfolio—the diversity of species grown and their relative acreage—provides additional predictive signal for aggregate loss risk.

This report documents an end-to-end empirical analysis testing that question. The study constructs a county×year panel spanning approximately 8,000–10,000 observations across the contiguous United States using three RMA administrative datasets. It then applies a sequence of analytical methods—from diversity index computation and spatial mapping, through non-parametric hypothesis tests and OLS regression, to ensemble machine learning with SHAP interpretability—to determine whether Shannon crop diversity is a meaningful, robust predictor of the insurance loss ratio.

1.1 Definition: The Loss Ratio

The primary outcome variable throughout this analysis is the **loss ratio**:

$$\text{Loss Ratio} = \frac{\text{Total Indemnity Paid}}{\text{Total Premium Earned}}$$

A loss ratio above 1.0 implies that indemnities exceed premiums collected—a net underwriting loss for insurers. This ratio is constructed *bottom-up* by summing indemnity and premium components from the sobsccc policy-summary table rather than averaging precomputed row-level ratios, avoiding unit-weighting bias.

2. Data Sources and Ingestion (Step 1)

2.1 Raw Data

Three RMA pipe-delimited text datasets are ingested:

Dataset	RMA Name	Description
colsommonth	Cost of Loss	Monthly indemnities by cause-of-loss code; each row is a county–crop–month–cause combination.
sobsccc	State/County/Crop	Policy-level summaries: premium, indemnity, liability, and coverage level by commodity and coverage category.
sobtpu	Type/Practice/Unit	Insured acreage by crop type, practice, and unit structure; used as the denominator for diversity calculations.

Files are read with `pandas.read_csv` using Latin-1 encoding (required for some historical RMA files) and column names assigned from the RMA data dictionaries. String columns are stripped of leading/trailing whitespace. FIPS state and county codes are zero-padded to 2 and 3 digits respectively throughout to ensure consistent merging.

2.2 Key Design Decision: Acres-Only Rows

Diversity metrics are computed exclusively from rows where `reporting_level_type == 'Acres'`. Rows reported in units of Trees, Head, or Tons are excluded. This ensures that crop proportions p_i reflect actual land-use shares and are dimensionally consistent across counties.

3. Diversity Index Construction (Step 2)

3.1 Crop Proportion Computation

For each county–year cell, the total insured acreage of crop i is divided by the county's total insured acreage to form the crop share p_i, with positive-acreage rows only:

$$p_i = \frac{a_i}{\sum_j a_j}, \quad p_i > 0$$

3.2 Diversity Metrics

Five complementary diversity metrics are computed from the $\{p_i\}$ distribution:

Metric	Formula	Interpretation
Species richness	$S = \#\{i : p_i > 0\}$	Raw count of distinct insured crops
Shannon entropy	$H' = -\sum_i p_i \ln p_i$	Combined richness and evenness; $H' = 0$ is monoc
Gini-Simpson index	$D = 1 - \sum_i p_i^2$	Probability that two random acres are different cr
Pielou's evenness	$J = H'/\ln S, \quad J \in [0, 1]$	How uniformly area is distributed; $J = 1$ is perfec
Berger-Parker dominance	$d = \max(p_i), \quad d \in [0, 1]$	Share of the single most-planted crop; $d = 1$ is mo

Shannon stability—the standard deviation of H' across years within a county—measures how much the crop mix changes over time, capturing a temporal dimension of resilience.

3.3 Output DataFrames

- `county_year_df.csv` — one row per county×year with all five metrics, insured acres, total indemnity, and derived per-acre figures.
- `county_summary.csv` — county-level averages across years, plus `shannon_stability` and `cv_indemnity` (coefficient of variation of indemnity/acre across years, measuring claim volatility).

4. Preliminary OLS Regression (Step 3)

As an initial sanity check, a simple bivariate OLS is estimated at the county level (one observation per county, averaged across all years):

$$\overline{\text{Indemnity/Acre}}_c = \alpha + \beta \cdot \overline{H'}_c + \varepsilon_c$$

Dixie County, FL is excluded as a structural outlier with anomalously high per-acre indemnities that would dominate the fit. The regression is run twice—on all counties and with Dixie excluded—to document outlier sensitivity. Both results are displayed with scatter plots overlaid with the OLS trend line, and R^2 and p-values are annotated directly on the figures.

5. Spatial Mapping (Steps 4–5)

5.1 Simpson Diversity Maps

County-level Simpson index values are merged to the 2025 Census TIGER/Line county shapefile using FIPS codes and plotted as choropleth maps—one PNG per year—using a consistent color scale anchored to the global min/max across all years to enable visual comparison. A special 2004 variant overlays the USDA ERS Farm Resource Region boundaries (obtained via `reglink.xls`) to contextualise diversity patterns within agro-climatic zones.

5.2 Corn Price Maps

A complementary set of maps shows corn gross value of production relative to the national average for each ERS Farm Resource Region from 2000 to 2024 (sourced from the USDA CornCostReturn dataset). Counties within each region are colored green if the regional price exceeds the national average and red if it falls below. These maps provide economic context for the crop composition patterns observed in the diversity maps.

6. K-Means Cluster Analysis (Step 6)

6.1 Algorithm: K-Means Clustering

K-Means partitions n counties into k non-overlapping clusters by minimising within-cluster sum of squared Euclidean distances in feature space:

$$\arg\min_{C_1,\ldots,C_k} \sum_{j=1}^{k} \sum_{x \in C_j} \|x - \mu_j\|^2$$

where μ_j is the centroid of cluster j. The algorithm iterates between:

1. **Assignment step**: assign each point to its nearest centroid.
2. **Update step**: recompute centroids as the mean of all assigned points.

until convergence. Ten random restarts (`n_init=10`) are used to reduce sensitivity to initialisation, and all features are standardised with `StandardScaler` (zero mean, unit variance) before clustering so that scale differences do not dominate.

6.2 Clustering Features

Six features are used: `mean_shannon`, `mean_evenness`, `mean_dominance`, `mean_richness`, `shannon_stability`, and `mean_acres` (county exposure size).

6.3 Choosing the Optimal k

The optimal number of clusters is selected by jointly examining two criteria across $k = 2, \dots, 9$:

- **Elbow method**: plot within-cluster inertia (SSE) vs. k; look for a kink where marginal gains from adding clusters diminish.
- **Silhouette score**: for each point, compute $s = (b - a)/\max(a, b)$ where a is mean intra-cluster distance and b is mean distance to the nearest other cluster. Scores range from -1 (misclassified) to $+1$ (well-separated); higher is better.

The k maximising the silhouette score is chosen as the best clustering.

6.4 Cluster Labelling and Profiling

Clusters are ordered by mean Shannon value (lowest → highest), relabelled "Cluster 1" (least diverse) through "Cluster k" (most diverse), and profiled by median diversity metrics, mean indemnity per acre, and coefficient of variation. A 2D PCA projection visualises cluster separation.

6.5 Random Forest Feature Importance

A Random Forest regressor (`n_estimators=300`) is trained within this step to predict `mean_indem_acre` from both diversity features and policy features (`mean_coverage_level`, `pct_buyup`, `mean_acres`, `diversity_category`). Feature importances (mean decrease in impurity, i.e., Gini importance) are plotted to compare diversity features against policy type and exposure features as predictors.

7. Volatility Regression (Step 7)

7.1 Motivation

Beyond the level of losses, insurers care about volatility—the year-to-year variability in indemnity payments. The outcome here is `cv_indemnity`, defined as:

$$\mathrm{CV} = \frac{\sigma(\text{indemnity}/\text{acre})}{\mu(\text{indemnity}/\text{acre})}$$

across all years for which a county has data.

7.2 Specification

A multiple OLS regression is estimated at the county level ($n \approx 2{,}600$):

$$\mathrm{CV}_c = \alpha + \beta_1 \cdot \overline{\text{richness}}_c + \beta_2 \cdot \overline{J}_c + \beta_3 \cdot \overline{d}_c + \beta_4 \cdot \overline{H'}_c + \beta_5 \cdot \sigma(H')_c + \varepsilon_c$$

7.3 Diagnostics

Variance Inflation Factors (VIF) are computed before fitting to flag multicollinearity. For a predictor X_j, VIF is:

$$\mathrm{VIF}_j = \frac{1}{1 - R_j^2}$$

where R_j^2 is the coefficient of determination from regressing X_j on all other predictors. VIF > 5 is flagged as a concern; VIF > 10 indicates severe multicollinearity. A 75/25 train/test split is used and both train and test R^2 and RMSE are reported to detect overfitting.

8. Cause-of-Loss Analysis by Cluster (Step 8)

Cause-of-loss codes from the colsommonth dataset are mapped to eight interpretable perils using RMA's official code list:

Group	RMA Codes Included
Drought	11, 13, 14
Heat	12, 22, 45
Cold	41, 42, 43, 44, 74
Wind/Storm	21, 61, 62, 63, 64
Precipitation	31, 51, 65, 67, 92
Pest	71, 73, 93
Disease	76, 80, 81, 82
Fire	91, 95

For each county×cause group pair, mean indemnity per acre (loss intensity) and CV across years (loss volatility) are computed. These are then aggregated to the cluster×cause group level (medians) and visualised as heatmaps.

8.1 Kruskal-Wallis Test

To formally test whether diversity cluster membership predicts volatility *within* each cause group, a Kruskal-Wallis H-test is applied:

$$H = \frac{12}{n(n+1)} \sum_{j=1}^{k} \frac{R_j^2}{n_j} - 3(n+1)$$

where R_j is the sum of ranks in group j. The test is a non-parametric analogue of one-way ANOVA, appropriate here because indemnity/acre distributions are heavily right-skewed. p-values are reported with stars ($* < 0.05$, $** < 0.01$, $*** < 0.001$).

9. County-Year Panel Construction (Step 9)

9.1 Panel Assembly

A comprehensive county×year panel is assembled from all three RMA tables plus county landmass data. Each row captures:

- From sobtpu: total insured acres, total indemnity, indemnity per insured acre.
- From colsommonth: indemnity and policy counts broken out by each of the eight cause groups; binary event flags indicating whether any claims occurred.
- From sobsccc: total policies earning premium, total liability, and the bottom-up loss ratio.
- From landmass: county land area in acres, percentage of land insured.

9.2 Loss Ratio Construction

Rather than averaging the pre-computed row-level `loss_ratio` column from sobsccc (which would be unit-weighted), the loss ratio is reconstructed as:

$$\mathrm{LR}_{cy} = \frac{\sum \mathrm{indemnity}_{cy}}{\sum \mathrm{premium}_{cy}}$$

using the full sobsccc table aggregated to county×year. This premium-weighted construction eliminates distortion from rows with very small premium bases.

9.3 Cause HHI and New Derived Variables

The Herfindahl-Hirschman Index (HHI) of cause-of-loss concentration is computed per county-year:

$$\text{HHI}_{cy} = \sum_{c} \left(\frac{\text{indemnity}_{cy,c}}{\text{indemnity}_{cy}} \right)^2$$

A value near 1.0 means one peril dominates; near zero means losses are spread across many causes. This complements crop-mix diversity with a measure of peril diversity.

10. Panel OLS Regressions (Step 10)

Two OLS models are estimated with 75/25 train/test splits:

Model 1 (county-year level, $n \approx$ 10,000+):

$$\text{indemnity/acre}_{cy} = \alpha + X_{cy}\beta + \gamma_1(\text{shannon} \times \text{pct\_insured}) + \gamma_2(\text{dominance} \times \text{acres}) + \gamma_3(n_{\text{crops}} \times \text{land\_acres}) + \varepsilon$$

Interaction terms are included to test whether diversity effects are heterogeneous by county size and agricultural intensity.

Model 2 (county level, $n \approx$ 2,600):

$$\text{CV(indemnity)}_c = \alpha + \bar{X}_c\beta + \varepsilon$$

where $\bar{X}$ denotes county means across years. This tests whether diversity predicts long-run volatility rather than single-year loss intensity.

VIF diagnostics are reported for both models. A side-by-side coefficient comparison plot highlights whether predictors have consistent signs across the intensity and volatility specifications.

11. Mega Correlation Matrix (Step 11)

A Spearman rank correlation matrix is computed across approximately 50 variables spanning all variable groups. Spearman correlation is preferred over Pearson because it is robust to the heavy right-skew of indemnity and loss-ratio distributions. Formally:

$$r_s(X, Y) = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$$

where d_i is the rank difference for observation i.

Axes are reordered by hierarchical clustering (Ward linkage on the distance matrix $1-|r_s|$) so that correlated variable blocks appear adjacent, using SciPy's `hierarchy.dendrogram`. Axis tick labels are color-coded by variable group (loss outcomes, crop diversity, cause diversity, exposure/scale, policy features, indemnity by cause, policies by cause, event flags) via a legend.

12. Feature Matrix (Step 12)

The analysis-ready feature matrix consolidates one row per county×year with the following final columns:

Feature	**Description**
`shannon_diversity`	Shannon entropy H'
`num_unique_crops`	Species richness S
`top_crop_share`	Berger-Parker dominance d
`evenness`	Pielou's J
`total_liability`	Total insured liability ($)
`avg_coverage_level`	Liability-weighted mean coverage level
`dominant_insurance_plan`	Plan with the most liability
`num_loss_causes`	Distinct cause-type groups with any claims
`weather_loss_share`	Fraction of indemnity from weather causes
`loss_ratio`	**Target**: indemnity / premium

Weather loss share is constructed so its numerator and denominator both come from the colsommonth table, keeping the ratio in $[0, 1]$ and avoiding contamination from sobtpu's different acreage grain.

13. Exploratory Data Analysis (Step 13)

Five figures are produced:

1. **Scatter plot**: Shannon diversity vs. loss ratio for the top 20 states by observation count, with an OLS trend line annotated with r, slope, and p-value. Loss ratio is capped at the 99th percentile to reduce distortion from audit outliers.
2. **Box plot**: Loss ratio distributions stratified by diversity quartile. Quartile cutpoints are computed on the full sample with `pd.qcut`, and median values are annotated on

each box.

3. **Correlation heatmap**: Pearson correlations among the nine feature-matrix variables, displayed as a lower-triangular heatmap (seaborn/coolwarm) with cell annotations.
4. **Choropleth maps**: Side-by-side county-level choropleths of mean Shannon diversity (YlGn palette) and mean loss ratio (RdYlGn_r palette), merged to the 2025 Census TIGER/Line shapefile. Missing counties are shown in light grey.
5. **Time series**: County-median Shannon diversity and loss ratio by year, with interquartile range bands, to examine whether the two series co-move over time.

14. Hypothesis Testing (Step 14)

14.1 Motivation and Test Selection

The Shapiro-Wilk test (applied to a 5,000-observation sample) strongly rejects normality of the loss ratio distribution (heavily right-skewed, kurtosis > 3). Non-parametric tests are therefore used throughout. Analyses are conducted at two levels:

- **County-year level**: $n \approx$ 10,000; note that the same county appears in multiple years (non-independent observations).
- **County average level**: $n \approx$ 2,700; one point per county, genuinely independent observations.

14.2 Test 1: Spearman Correlation

Spearman's r_s is computed between each of the four diversity metrics (`shannon_diversity`, `num_unique_crops`, `top_crop_share`, `evenness`) and the loss ratio, at both levels of analysis. Effect sizes are the r_s values themselves; p-values are adjusted using SciPy's two-sided approximation.

14.3 Test 2: Kruskal-Wallis ANOVA

The sample is divided into diversity quartiles and the Kruskal-Wallis H-statistic is computed to test the null that all quartiles share the same loss ratio distribution:

$$H_0 : F_1(x) = F_2(x) = F_3(x) = F_4(x)$$

Effect size is reported as:

$$\eta^2 = \frac{H - k + 1}{n - k}$$

Pairwise Mann-Whitney U tests between all $\binom{4}{2} = 6$ quartile pairs are then run with Bonferroni correction (multiplying p-values by 6) to control the family-wise error rate.

14.4 Test 3: Mann-Whitney U Test

Counties are split at the median of `shannon_diversity` into a high-diversity and low-diversity group and a two-sided Mann-Whitney U test is applied:

$$U = n_1 n_2 + \frac{n_1(n_1+1)}{2} - R_1$$

where R_1 is the sum of ranks in group 1. Effect size is reported as the rank-biserial correlation $r = 1 - 2U/(n_1 n_2) \in [-1, 1]$.

15. OLS Regression Models (Step 15, Stage A)

15.1 Target Variable Transformation

The loss ratio is heavy-tailed and includes zeros, so the target is transformed:

$$y = \log(1 + \text{loss\_ratio})$$

This compresses the right tail and ensures that zero-loss observations remain in the sample. Coefficients β on log-transformed targets are interpreted as approximately $100\beta\%$ changes in the loss ratio for a one-unit change in the predictor.

15.2 Three Nested Models

Model A1 — Diversity only (4 predictors):

$$\log(1 + \text{LR}) = \alpha + \beta_1 H' + \beta_2 S + \beta_3 d + \beta_4 J + \varepsilon$$

Model A2 — With controls (8 predictors):

$$\log(1 + \text{LR}) = \alpha + \boldsymbol{\beta}_{\text{div}} \cdot \mathbf{x}_{\text{div}} + \beta_5 \log(\text{liability}) + \beta_6 \bar{\ell} + \beta_7 N_c + \beta_8 w + \varepsilon$$

where $\bar{\ell}$ is liability-weighted mean coverage level, N_c is the number of distinct loss causes, and w is the weather loss share.

Model A3 — Year and state fixed effects, clustered SE:

$$\log(1 + \text{LR}) = \alpha + \boldsymbol{\beta}_{\text{div}} \cdot \mathbf{x}_{\text{div}} + \boldsymbol{\beta}_{\text{ctrl}} \cdot \mathbf{x}_{\text{ctrl}} + \delta_t + \gamma_s + \varepsilon_{cty}$$

State fixed effects (γ_s) absorb time-invariant geographic confounders (soil type, climate zone). Year fixed effects (δ_t) absorb common shocks (e.g., the 2002 drought). Standard errors are clustered at the county level to correct for within-county serial correlation of errors.

15.3 Temporal Train/Test Split

The data are split by year rather than by random sampling:

- **Train**: 1999, 2000, 2001
- **Test**: 2004 (out-of-sample, held out entirely during estimation)

This mimics real actuarial use: a model estimated on past data is evaluated on genuinely future observations. For Model A3 (which includes year fixed effects), the 2004 test observations are scored using the 2001 year intercept as the nearest training-year proxy.

15.4 Key Results

Model	**Train R^2**	**Test R^2 (2004)**	$\hat{\beta}_{H'}$	**SE**	**Sig.**
A1 – Diversity only	0.043	−0.009	0.506	0.049	***
A2 – + Controls	0.217	0.092	0.691	0.049	***
A3 – + Year & State FE	0.298	0.048	0.396	0.058	***

The positive sign on $\hat{\beta}_{H'}$ is counterintuitive at first glance but consistent across all three specifications: *conditional on the other regressors*, higher Shannon diversity is associated with a higher log loss ratio. The interpretation is nuanced: `top_crop_share` (dominance) enters the same regression with a large positive coefficient ($\hat{\beta} \approx 1.2$–1.3), creating a multicollinearity structure where H' and d together account for the diversity-risk relationship. More diverse counties—holding coverage level and weather exposure constant—may face more complex exposure structures that translate to higher gross loss ratios.

16. Gradient Boosting and SHAP (Step 15, Stage B)

16.1 Algorithm: Gradient Boosting Regression

Gradient Boosting builds an ensemble of decision trees sequentially. Each tree f_m is trained to predict the *negative gradient* (pseudo-residual) of the loss function L with respect to the current ensemble prediction:

$$r_{im} = -\left[\frac{\partial L(y_i, F(x_i))}{\partial F(x_i)}\right]_{F=F_{m-1}}$$

For squared-error loss, this simplifies to the residual $y_i - F_{m-1}(x_i)$. The ensemble prediction is updated:

$$F_m(x) = F_{m-1}(x) + \nu \cdot f_m(x)$$

where $\nu \in (0, 1]$ is the learning rate (shrinkage parameter). A smaller ν requires more trees but typically achieves better generalisation. The final model sums all trees:

$$\hat{y} = F_M(x) = F_0(x) + \nu \sum_{m=1}^{M} f_m(x)$$

Scikit-learn's `GradientBoostingRegressor` is used. A complementary `RandomForestRegressor` (bagging-based ensemble) is also trained for comparison.

16.2 Temporal Train/Test Split

The same year-based split as Stage A is used (train: 1999–2001; test: 2004), ensuring that predictive performance reflects genuine out-of-sample generalisation.

16.3 SHAP Values

SHAP (SHapley Additive exPlanations) decomposes a model's prediction into additive contributions from each feature, grounded in cooperative game theory. For a prediction $F(x)$ and feature j:

$$\phi_j = \sum_{S \subseteq \mathcal{F} \setminus \{j\}} \frac{|S|!\,(|\mathcal{F}| - |S| - 1)!}{|\mathcal{F}|!} \left[F_{S \cup \{j\}}(x) - F_S(x)\right]$$

where the sum is over all subsets S of features excluding j, and F_S is the model re-evaluated using only features in S. SHAP values satisfy:

- **Efficiency**: $\sum_j \phi_j = F(x) - \mathbb{E}[F(x)]$.
- **Symmetry**: equal contribution features receive equal SHAP values.
- **Dummy**: a feature that never changes the prediction has $\phi_j = 0$.

Two visualisations are produced:

- **Beeswarm summary**: one dot per observation, x-axis = SHAP value, colored by raw feature value. Reveals how the direction and magnitude of each feature's effect varies across the sample.
- **Dependence plots**: SHAP value of Shannon diversity vs. its raw value, colored by top-crop share, to detect interaction effects.

16.4 Partial Dependence Plots

Partial Dependence Plots (PDPs) show the marginal effect of a feature after averaging out all other features:

$$\hat{f}_j(x_j) = \mathbb{E}_{\mathbf{x}_{\setminus j}}\left[F(x_j, \mathbf{x}_{\setminus j})\right] \approx \frac{1}{n}\sum_{i=1}^{n} F(x_j, x_{i,\setminus j})$$

PDPs for `shannon_diversity` and `top_crop_share` are produced to complement the SHAP analysis.

17. Robustness Checks (Step 16)

Four potential confounds are tested against the benchmark Model A3.

17.1 Check 1: Geography Confound

The 50 state fixed effects are replaced with 9 ERS Farm Resource Region fixed effects (agro-climatic zones that better capture soil type, typical crop mix, and rainfall patterns). If the diversity signal is merely a geographic proxy for the Corn Belt or Great Plains, it should vanish under region controls. Within-region Spearman correlations are also computed to verify the relationship holds inside each zone.

17.2 Check 2: Scale Confound

Species richness (S) grows mechanically with county size—larger counties can support more crops. Shannon entropy and Pielou's evenness are scale-invariant (they depend only on proportions), but richness and dominance are not. This check runs OLS using only scale-invariant metrics, and repeats the analysis within quartiles of insured acres to confirm that the diversity effect is not driven by county size.

17.3 Check 3: Commodity Composition Confound

Crop choice itself may predict expected loss ratios independently of diversity. Corn and soybeans, for example, have different historical loss profiles than cotton or specialty crops. Acreage shares of the top commodities (corn, soybeans, wheat, cotton, vegetables, hay) are computed from sobtpu and added as controls. A "crop-mix-adjusted" residual is constructed to test whether Shannon diversity explains variance beyond what crop-level shares account for.

17.4 Check 4: Coverage Selection Bias

If high-risk counties purchase more crop insurance (adverse selection), the loss ratio denominator is inflated in risky counties, potentially confounding the diversity signal. The percentage of county land area under insurance coverage (`pct_land_insured`) is added as a control. If selection drives the main results, Shannon diversity should lose significance after this addition.

17.5 Coefficient Stability Plot

A forest plot displays the Shannon diversity coefficient $\hat{\beta}_{H'}$ and its 95% confidence interval across all nine specifications (baseline A3 plus four check variants plus sub-group analyses). Coefficient stability across specifications is the primary evidence of robustness.

18. Summary of Findings

Analysis	Key Finding
Spearman correlation	Shannon diversity is significantly positively correlated with loss ratio at both the county-year and county-average levels (r_s reported at both levels; all $p < 0.001$).
Kruskal-Wallis	Loss ratio distributions differ significantly across diversity quartiles; effect size η^2 quantifies the proportion of rank variance explained.
OLS (A1–A3)	Shannon diversity retains a positive, statistically significant coefficient across all three nested models, including after adding state and year fixed effects with clustered standard errors.
Gradient Boosting + SHAP	SHAP analysis reveals the non-linear, interaction-dependent structure of diversity's effect on predicted loss ratios, complementing the linear OLS results.
Robustness checks	Coefficient sign and significance are stable across geography, scale, commodity composition, and coverage selection controls.
Cause-of-loss cluster analysis	Diversity clusters differ in loss intensity and volatility within individual peril categories (Kruskal-Wallis significant for most causes).

19. Limitations and Caveats

1. **Positive diversity coefficient**: The counterintuitive positive association between H' and loss ratio (controlling for d and other features) warrants further investigation. It may reflect that counties with more diverse *insured* crops face more diverse risk profiles that the current actuarial premiums do not fully price, rather than that diversity itself causes higher losses.
2. **Panel dependence**: The county-year panel contains repeated observations from the

same counties. OLS standard errors (even when clustered at the county level) may not fully account for spatial autocorrelation between neighbouring counties.

3. **Temporal coverage**: The training years (1999–2001) and test year (2004) span a period that includes the 2002 drought—one of the most severe in recent US agricultural history. Results may not generalise to post-2010 periods with different weather patterns and crop insurance program structures.
4. **Selection into insurance**: Counties with no insured acreage in a given year do not appear in the data. If uninsured counties differ systematically in diversity or risk, the analysis applies only to the insured population.
5. **Multicollinearity**: Shannon entropy, Pielou's evenness, and Berger-Parker dominance are algebraically related. High VIFs for these metrics limit the interpretability of individual coefficients in regression models that include all three simultaneously.

20. Computational Pipeline Summary

The analysis is implemented as a 16-step Python pipeline:

Step	Script	Output
0	`read_crop_loss_api.py`	Raw RMA .txt files
1	`01_load_data.py`	colsommonth.csv, sobsccc.csv, sobtpu.csv
2	`02_build_diversity.py`	county_year_df.csv, county_summary.csv
3	`03_regression.py`	Bivariate OLS plots
4	`04_cluster_analysis.py`	Clustered county_summary.csv, RF importances
5	`05_maps_simpson.py`	simpson_maps/*.png
6	`06_maps_corn_price.py`	corn_price_maps/*.png
7	`07_volatility_regression.py`	volatility_regression.png
8	`08_cause_of_loss_by_cluster.py`	cause_of_loss_heatmaps.png
9	`09_county_panel.py`	county_panel.csv
10	`10_panel_regression.py`	Panel OLS figures
11	`11_mega_correlation.py`	mega_corr_matrix.png
12	`12_feature_matrix.py`	feature_matrix.csv
13	`13_eda.py`	eda_plots/*.png
14	`14_hypothesis_tests.py`	hypothesis_tests/*.png, results_summary.txt
15	`15_modeling.py`	OLS summaries, GBM/RF figures, SHAP plots
16	`16_robustness.py`	Robustness figures, stability_table.csv

Key Python libraries: `pandas`, `numpy`, `scipy`, `statsmodels`, `scikit-learn`, `shap`, `matplotlib`, `seaborn`, `geopandas`.